

Aryaman Bahukhandi

Berkeley, CA | ✉ abahukhandi@ucdavis.edu | 🌐 Portfolio | 🔗 LinkedIn | 🐙 GitHub

SUMMARY

Software Engineer with expertise in backend engineering, agentic systems, and MLOps/DevOps automation. Proven ability to deliver scalable and efficient systems, reduce latency, and optimize cloud infrastructure. Passionate about AI-driven applications and building production-ready software with real-world impact.

EDUCATION

- **Master of Science in Computer Science** September 2022 – June 2025
University of California, Davis Major GPA: 3.81
- **Bachelors of Engineering in Computer Engineering** August 2018 – July 2022
Pune Institute of Computer Technology GPA: 3.85

EXPERIENCE

- **Stealth Startup (AI/Healthcare)** August 2025 – Present
MLOps Engineer Davis, CA
 - Led MLOps for **Inquisito**, an agentic platform for FDA premarket submission document generation and editing.
 - Deployed containerized **FastAPI/LangGraph** microservices on **AWS ECS & EKS** with **Kubernetes, Docker**, and **GitHub Actions** CI/CD, achieving **99.9% uptime**.
 - Accelerated **PubMedBERT** embedding inference using **TensorRT** and **ONNX Runtime** with **FP16** optimization, **improving throughput by 2×** and reducing latency across **RAG** pipelines for FDA submission retrieval and ranking.
 - Automated ingestion and indexing of FDA documents using **Python & Bash** scripts, **AWS EC2** for compute and **S3** for storage. Cut **index load times to <10s**, achieving SLOs of **<50ms semantic search** and **<1s agentic workflows**.
 - Implemented observability with **CloudWatch** dashboards, and **LangSmith** tracing, cutting incident triage time by **35%**.
- **Exploratory Systems Lab** July 2025 – September 2025
Software Engineer Davis, CA
 - Designed and implemented a **Retrieval-Augmented Generation (RAG)** agent on **ResilientDB**, a high-throughput open-source **blockchain**, to support QA tasks for exploring documentation and setup suggestions for client deployments.
 - Utilized **LlamaIndex** for document parsing and semantic chunking, stored in **PostgreSQL (pgvector extension)**.
 - Achieved **30% faster retrieval** and concurrent semantic queries at **sub-50 ms latency** for **100k+ embedded chunks**.
 - Built production ready full-stack services using **TypeScript, Next.js & React** with **pgvector** queries.
- **Visualization And Interface Design Innovation Lab** May 2023 – May 2025
ML Research Engineer Davis, CA
 - Engineered the evaluation framework for **Awesum**, a visual analytics system for prompt evaluation in summarization.
 - Developed custom intelligent agents for feature recommendation and prompt suggestions using **OpenAI's GPT-3.5-Turbo**.
 - Integrated **LangChain & LangGraph** to manage and track multi-turn conversations between users and AI agents.
 - Implemented **concurrent OpenAI API calls**, with **exponential backoff, wait queues, and token-limiting** strategies to optimize throughput and **reduce dataset processing latency by 80%**.
- **Huf Group** September 2021 – February 2022
Software Engineer Intern Pune, India
 - Collaborated with cross-functional teams to deploy a full-stack **SaaS** platform for digitizing manufacturing operations.
 - Developed **RESTful JavaScript** back-end services (**Node.js, Express**) on **AWS EC2** with a **MongoDB** data layer.
 - Designed **React** dashboards to display real-time machine metrics, enhancing operational visibility.
 - Improved API performance by **45%**, increasing daily manufacturing throughput by **15%**.
- **College of Military Engineering** March 2021 – August 2021
Software Engineer - DevOps Intern Pune, India
 - Developed **Python microservices** for an inventory system, boosting graph processing latency by **50%** & **2x throughput**.
 - Built a fault-tolerant data pipeline with priority-based queuing using **Apache Kafka**.
 - Engineered **Docker & Kubernetes** based containerization & cluster orchestration on **ECS,EKS**, achieving **99% uptime**.
 - Set up **CI/CD** with **GitHub Actions** and **Bash** scripts for **unit and integration tests**, reducing deployment time by **40%**.

TECHNICAL SKILLS

Languages: C++, Python, Bash, Java, JavaScript, TypeScript, HTML, CSS

Frameworks/Libraries: React, Next, Node, Flask, FastAPI, REST, TensorFlow, PyTorch, LangChain, LlamaIndex

DevOps/Cloud: Google Cloud, AWS, Docker, Kubernetes, Helm, Kafka, Git, GitHub Actions, Terraform

Databases/OS: PostgreSQL, MySQL, SQLite, MongoDB, Redis, Qdrant, LINUX, UNIX

Relevant Skills: MLOps/DevOps, Full-Stack Development, Microservices, Serverless Architecture, Cloud Infrastructure, Containerization & Orchestration, Agile Development, Web Development, CI/CD, System Design, API Design

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Database Management Systems, Computer Architecture, Distributed Systems, Big Data & Statistics, Machine Learning, Software Engineering.